

Notes on

Partial Differential Equations

Lecturers
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This text consists of notes on the course Partial Differential Equations, lectured by Professor Alaka Das (AD) and Professor Sudeshna Banerjea (SB) at Jadavpur University in the 5th semester 2025.

Some changes and some additions have been made by me. To distinguish them from the lecture's actual contents, they are labelled with an asterisk. So any *Lemma** or *Remark** or *Proof** that the reader might come across are wholly my responsibility.

Please report errors, typos etc. to this email: dassayan0013@gmail.com.

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List of Lectures

Lecture 1 (8th August, 2025)	1
(AD) First order PDEs: origins, general solution, constructions.	
Lecture 2 (5th August, 2025)	2
(SB) Second order PDEs: origins, 3 main questions, classification (linear, semi-linear, quasi-linear); the Cauchy problem: condition for uniqueness and Cauchy-Kovalevskaya theorem, characteristic curve.	
Lecture 3 (12th August, 2025)	2
(SB) Reduction to canonical form, classification (elliptic, parabolic, hyperbolic) and characteristic curves.	

Course organisation. — The lectures will take place on Tuesdays from from 10:00 to 12:00 and on Fridays from 11:00 to 13:00, in the UG-III classroom on the ground floor of Ramanujan Bhavan.

The prerequisite to this lecture is an undergraduate course on Ordinary Differential Equations, akin to the one covered in the fourth semester course [Ordinary Differential Equations](#). For this course, the references are Ian Sneddon's *Elements of PDEs* ([\[Sne06\]](#)) as well as Tyn Myint-U and Lokenath Debnath's *Linear PDEs* ([\[MD07\]](#)).

The syllabus of this course is available [here](#).

CHAPTER 1.

First order PDEs

1

1.1. Introduction

LECTURE 1 1.1.1. Origins of first order PDEs. —
8th Aug, 2025

CHAPTER 2.

Second order PDEs

2

2.1. The Cauchy problem

LECTURE 2
5th Aug, 2025

2.1.1. Origins of second order PDEs. —

2.1.2. The Cauchy problem. —

2.2. Reduction to canonical form and classification

LECTURE 3
12th Aug, 2025

2.2.1. Canonical form. —

Bibliography

- [MD07] Tyn Myint-U and Lokenath Debnath. *Linear Partial Differential Equations for Scientists and Engineers*. Birkhäuser, 2007.
- [Sne06] Ian N. Sneddon. *Elements of Partial Differential Equations*. Dover Books on Mathematics. Dover Publications, 2006.